



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

Semester II 2019/2020

Subject : Bioinformatics I (SCSB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 05 – Advanced Database Searching

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- 1) Create an artificial protein sequence consisting of human RBP4 followed by the C2 domain of human protein kinase C α . An example of this is shown in Web Document 5.5 (available in your e-Learning site). Enter this combined sequence into a PSI-BLAST search
 - i) In general, are multiple domains always detected by the PSI-BLAST program?
 - ii) Do any naturally occurring proteins have both lipocalin and C2 domains?

First, find RBP4 Homo sapiens and select NP_006735.2 the copy the FASTA sequence.

```
>NP_006735.2 retinol-binding protein 4 isoform a precursor [Homo sapiens]
```

```
MKWWWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFSVDETGQ
```

```
MSATAKGRVRLNNWDVCADMVGTFDTEDPAKFKMKYWGVASFLQKGNDHIVDTDYDTYAVQYSCRL
```

```
LNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLL
```

retinol-binding protein 4 isoform a precursor [Homo sapiens]

NCBI Reference Sequence: NP_006735.2

[GenPept](#) [Identical Proteins](#) [Graphics](#)

```
>NP_006735.2 retinol-binding protein 4 isoform a precursor [Homo sapiens]
```

```
MKWWWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFSVDETGQ
```

```
MSATAKGRVRLNNWDVCADMVGTFDTEDPAKFKMKYWGVASFLQKGNDHIVDTDYDTYAVQYSCRL
```

```
LNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLL
```

Then find Protein kinase C alpha Homo sapiens and select NP_002728.2 the copy the FASTA sequence.

>NP_002728.1 protein kinase C alpha type [Homo sapiens]

MADVFPGNDSTASQDVANRFARKGALRQKNVHEVKDHKFIARFFKQPTFCSHCTDFIWGFGKQGFQCQVC
CFVVHKRCHEFVTFSCPGADKGPDTDDPRSKHKFKIHTYGSPTFCDHCGSLLYGLIHQGMKCDTCDMNVH
KQCVINVPSLCGMDHTEKRGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQ
KTKTIRSTLNPQWNESFTFKLKPSDKDRRLSVEIWDWDRTRNDFMGSLSFGVSELMKMPASGWYKLLNQ
EEGEYYNVPPIEGDEEGNMELRQKFEKAKLGPAGNKVISPSEDRKQPSNNLDRVKLTDFNFLMVLGKGSF
GKVMLADRKGTEELYAIKILKKDVVIQDDDVECTMVEKRVLALLDKPPFLTQLHSCFQTVDRLYFVMEYV
NGGDLMYHIQQVGKFKEPQAVFYAAEISIGLFFLHKRGIYRDLKLDNVMLDSEGHKIADFGMCKEHMM
DGVTTTRTCGTPDYIAPEIIAYQPYGKSVDWWAYGVLLYEMLAGQPPFDGEDEDELFSIMEHNVSPYKS
LSKEAVSICKGLMTKHPAKRLGCGPEGERDVREHAFFRRIDWEKLENREIQPPFKPKVCGKGAENFDKFF
TRGQPVLTPPDQLVIANIDQSDFEFVSYPNPQFVHPILQSAV

[FASTA](#) ▾

protein kinase C alpha type [Homo sapiens]

NCBI Reference Sequence: NP_002728.1

 **This sequence has been updated. [See current version.](#)**

[GenPept](#) [Identical Proteins](#) [Graphics](#)

>NP_002728.1 protein kinase C alpha type [Homo sapiens]
MADVFPGNDSTASQDVANRFARKGALRQKNVHEVKDHKFIARFFKQPTFCSHCTDFIWGFGKQGFQCQVC
CFVVHKRCHEFVTFSCPGADKGPDTDDPRSKHKFKIHTYGSPTFCDHCGSLLYGLIHQGMKCDTCDMNVH
KQCVINVPSLCGMDHTEKRGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQ
KTKTIRSTLNPQWNESFTFKLKPSDKDRRLSVEIWDWDRTRNDFMGSLSFGVSELMKMPASGWYKLLNQ
EEGEYYNVPPIEGDEEGNMELRQKFEKAKLGPAGNKVISPSEDRKQPSNNLDRVKLTDFNFLMVLGKGSF
GKVMLADRKGTEELYAIKILKKDVVIQDDDVECTMVEKRVLALLDKPPFLTQLHSCFQTVDRLYFVMEYV
NGGDLMYHIQQVGKFKEPQAVFYAAEISIGLFFLHKRGIYRDLKLDNVMLDSEGHKIADFGMCKEHMM
DGVTTTRTCGTPDYIAPEIIAYQPYGKSVDWWAYGVLLYEMLAGQPPFDGEDEDELFSIMEHNVSPYKS
LSKEAVSICKGLMTKHPAKRLGCGPEGERDVREHAFFRRIDWEKLENREIQPPFKPKVCGKGAENFDKFF
TRGQPVLTPPDQLVIANIDQSDFEFVSYPNPQFVHPILQSAV

Here is the C2 domain of PKCA, obtained by clicking the C2 “region” link:

>NP_002728.1:159-289 protein kinase C alpha type [Homo sapiens]

RGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQKTKTIRSTLNPQWNESFT
FKLKPSDKDRRLSVEIWDWDRTRNDFMGSLSFGVSELMKMPASGWYKLLNQEEGEYYNVP

FASTA ▾

protein kinase C alpha type [Homo sapiens]

NCBI Reference Sequence: NP_002728.1

⚠ **This sequence has been updated. [See current version.](#)**

[GenPept](#) [Identical Proteins](#) [Graphics](#)

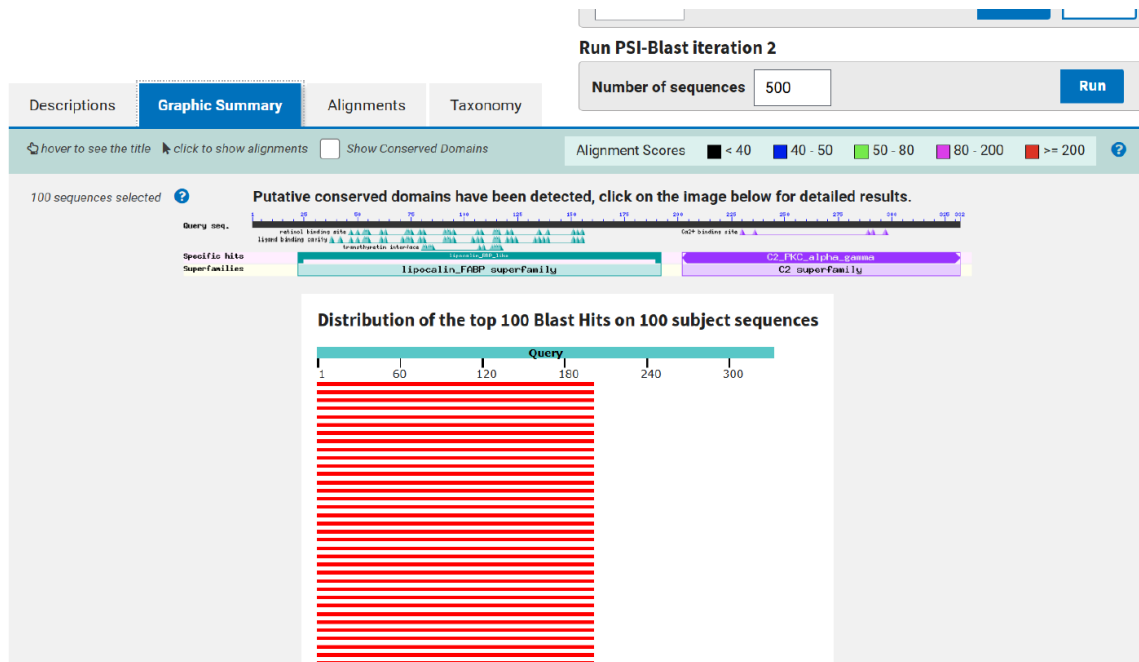
```
>NP_002728.1:159-289 protein kinase C alpha type [Homo sapiens]
RGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQKTKTIRSTLNPQWNESFT
FKLKPSDKDRRLSVEIWDWDRTRNDFMGSLSFVSELMKMPASGWYKLLNQEEGEYYNVP
```

Here is a chimeric protein:

```
MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFSVDETQMSA
TAKGRVRLNNWDVCADMGVTFDTEDPAKFKMKYWGVASFLQKGNDDHWIVDTDYDTYAVQYSCRLNLNDG
TCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLLRGRIYLKAEVADEKLHVTVR
DAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQKTKTIRSTLNPQWNESFTFKLKPSDKDRRLSVEIWDWDRTRND
FMGSLSFVSELMKMPASGWYKLLNQEEGEYYNVP
```

The screenshot shows the NCBI BLAST web interface. The top navigation bar includes the NCBI logo and links for Home, Recent Results, Saved Strategies, and Help. The main heading is "BLAST® » blastp suite". Below this, there are tabs for different BLAST programs: blastn, **blastp**, blastx, tblastn, and tblastx. The "Standard Protein BLAST" section is active. The "Enter Query Sequence" section contains a text area with the protein sequence, a "Query subrange" section with "From" and "To" fields, and a "Job Title" field. The "Choose Search Set" section has a "Database" dropdown set to "Non-redundant protein sequences (nr)", an "Organism" dropdown, and an "Exclude" section with checkboxes for "Models (XM/XP)", "Non-redundant RefSeq proteins (WPI)", and "Uncultured/environmental sample sequences". The "Program Selection" section has a "Algorithm" dropdown set to "PSI-BLAST (Position-Specific Iterated BLAST)". A "BLAST" button is at the bottom left. A note at the bottom states: "Note: Parameter values that differ from the default are highlighted in yellow and marked with * sign". The bottom status bar shows the system clock as 5°C sunny, a search bar, and the date/time as 8:46 AM 30/4/2025.

This shows the PSI-BLAST result:



The graphical portion of the result suggests that there are matches to either RBP4 or to PKC but not both; the RBP4 portion of the protein is larger so it accumulates higher scores (and lower E values) and those results are listed first. By inspection there are no proteins with both domains.

2) The purpose of this problem is to compare BLASTP to DELTA-BLAST

The malarium parasite *Plasmodium vivax* has a multigene family called vir that is specific to that organism (del Portillo et al., 2001). There are 600–1000 copies of these genes, and they may have a role in causing chronic infection through antigenic variation.

- i) Select vir1 and perform a BLASTP search of the nonredundant protein database (restricting the species to *Plasmodium vivax*).
- ii) Then perform a DELTA-BLAST search with the same entry. For each search, approximately how many proteins have an E value less than $1e-10$?

First , find vir-1 protein from *Plasmodium vivax* search NCBI Protein: "*Plasmodium vivax*"[ORGN] vir1 , then enter its query sequence CAB96690.1

BLAST® » blastp suite Home Recent Results Saved Strategies Help

blastn **blastp** blastx tblastn tblastx Standard Protein BLAST

BLASTP programs search protein databases using a protein query, more...

[Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), g(i)s, or FASTA sequence(s) [Clear](#)

CAB96690.1 Query subrange: From To

Or, upload file Choose File No file chosen

Job Title CAB96690.virt [Plasmodium vivax]

Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database ☒ Standard databases (nr etc.) ☐ Experimental databases

Non-redundant protein sequences (nr)

Organism Optional Plasmodium vivax (taxid:5855) ☐ exclude [Add organism](#)

Exclude Optional ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

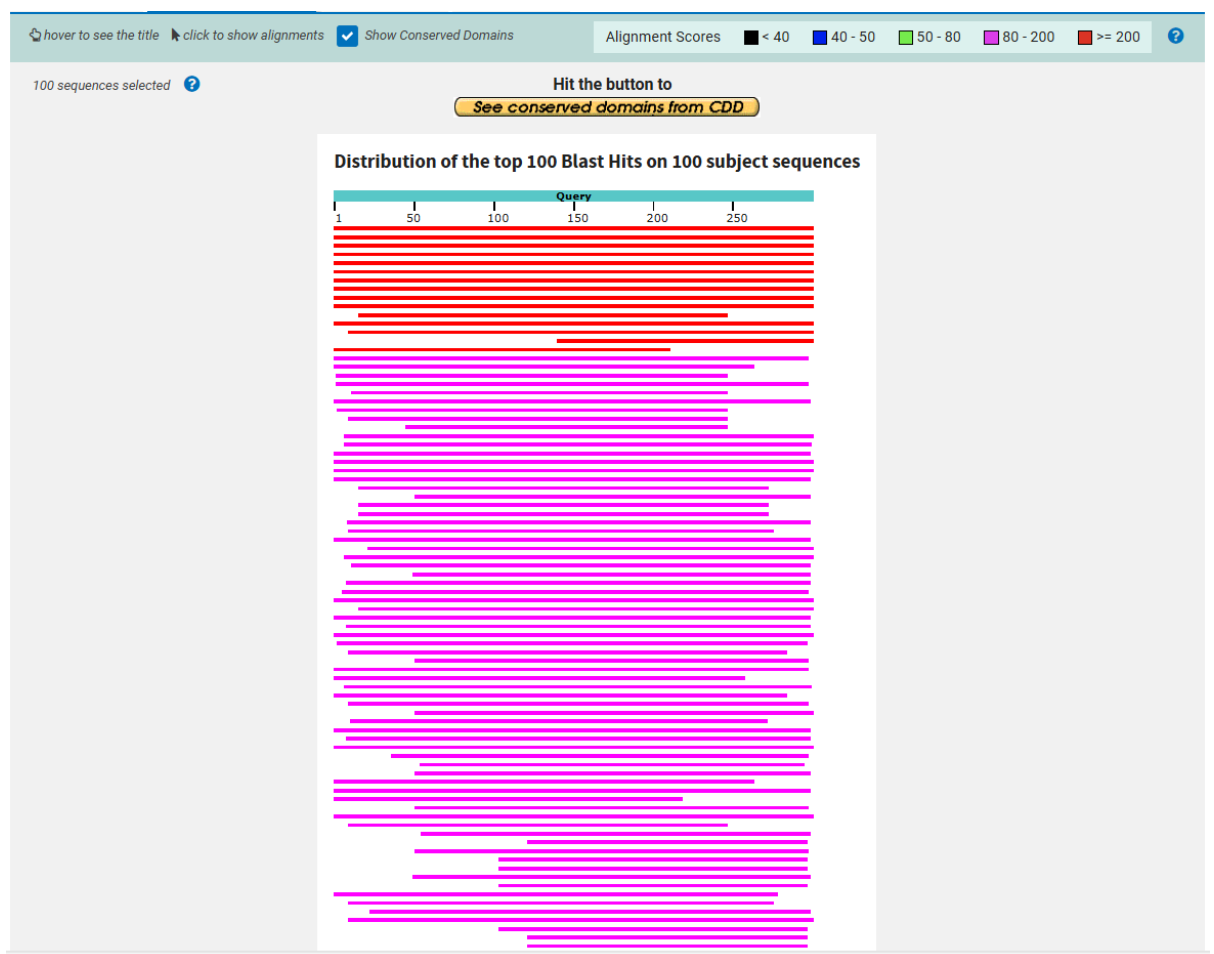
BLAST Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

[+ Algorithm parameters](#)

30°C Partly sunny Search 2:32 PM 30/4/2025

Here is the BLASTP result:



Descriptions	Graphic Summary	Alignments	Taxonomy					
Sequences producing significant alignments								
Download Select columns Show 100 ?								
☒ select all 100 sequences selected								
GenPept Graphics Distance tree of results Multiple alignment MSA Viewer								
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ vir1 [Plasmodium vivax]	Plasmodium vivax	607	607	100%	0.0	100.00%	299	CAB96690.1
☒ Plasmodium vivax Vir protein, putative [Plasmodium vivax]	Plasmodium vivax	535	535	100%	0.0	88.63%	299	SCA82045.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	516	516	100%	0.0	84.62%	299	CAI7724112.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	510	510	100%	0.0	85.28%	298	CAI7719847.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	494	494	100%	3e-178	83.61%	299	VUZ93672.1
☒ VIR protein [Plasmodium vivax]	Plasmodium vivax	488	488	100%	8e-176	82.27%	299	SCA60058.1
☒ VIR protein [Plasmodium vivax]	Plasmodium vivax	481	481	100%	4e-173	80.27%	299	SCA60268.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	423	423	100%	4e-150	72.58%	298	VUZ99642.1
☒ unnamed protein product [Plasmodium vivax]	Plasmodium vivax	416	416	100%	3e-147	69.90%	299	CAG9485322.1
☒ unnamed protein product [Plasmodium vivax]	Plasmodium vivax	396	396	100%	1e-139	66.22%	299	CAG9473090.1
☒ hypothetical protein PVMG_05816 [Plasmodium vivax Mauritania I]	Plasmodium vivax Mauritania I	389	389	77%	2e-137	84.78%	244	KMZ95672.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	377	377	100%	3e-131	58.38%	334	VUZ99734.1
☒ hypothetical protein PVNG_01538 [Plasmodium vivax North Korean]	Plasmodium vivax North Korean	345	345	97%	3e-119	58.17%	307	KNA02384.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	274	274	54%	6e-93	71.79%	195	VUZ99735.1
☒ hypothetical protein PVIIG_06431, partial [Plasmodium vivax India VII]	Plasmodium vivax India VII	267	267	70%	6e-90	65.24%	210	KMZ77291.1
☒ VIR protein [Plasmodium vivax]	Plasmodium vivax	189	189	99%	5e-58	41.47%	301	SCA81956.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	177	177	88%	4e-54	41.13%	264	VUZ99426.1
☒ VIR protein [Plasmodium vivax]	Plasmodium vivax	160	160	82%	2e-47	42.11%	248	SCA83617.1
☒ unnamed protein product [Plasmodium vivax]	Plasmodium vivax	157	157	99%	1e-45	37.58%	301	CAG9474536.1
☒ hypothetical protein PVBG_04826 [Plasmodium vivax Brazil I]	Plasmodium vivax Brazil I	149	149	78%	3e-43	40.51%	264	KMZ88617.1
☒ PIR protein [Plasmodium vivax]	Plasmodium vivax	135	135	99%	2e-37	33.12%	292	VUZ99362.1
☒ hypothetical protein PVMG_05324 [Plasmodium vivax Mauritania I]	Plasmodium vivax Mauritania I	129	129	81%	2e-35	34.15%	266	KMZ95725.1
☒ hypothetical protein PVNG_03824 [Plasmodium vivax North Korean]	Plasmodium vivax North Korean	129	129	79%	2e-35	35.56%	266	KMZ98985.1

Next “Edit and resubmit” and change from BLASTP to DELTA-BLAST.

BLAST® » blastp suite

HomeRecent ResultsSaved StrategiesHelp

blastnblastpblastxtblastntblastx

Standard Protein BLAST

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

Query subrange [?](#)

From

To

Or, upload file

No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database
 ☒ Standard databases (nr etc.)
 ☐ Experimental databases

[?](#)

Organism

☐ exclude
 [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude

☐ Models (XM/XP)
 ☐ Non-redundant RefSeq proteins (WP)
 ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)
 ☐ blastp (protein-protein BLAST)
 ☐ PSI-BLAST (Position-Specific Iterated BLAST)
 ☐ PHI-BLAST (Pattern Hit Initiated BLAST)
 ☒ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

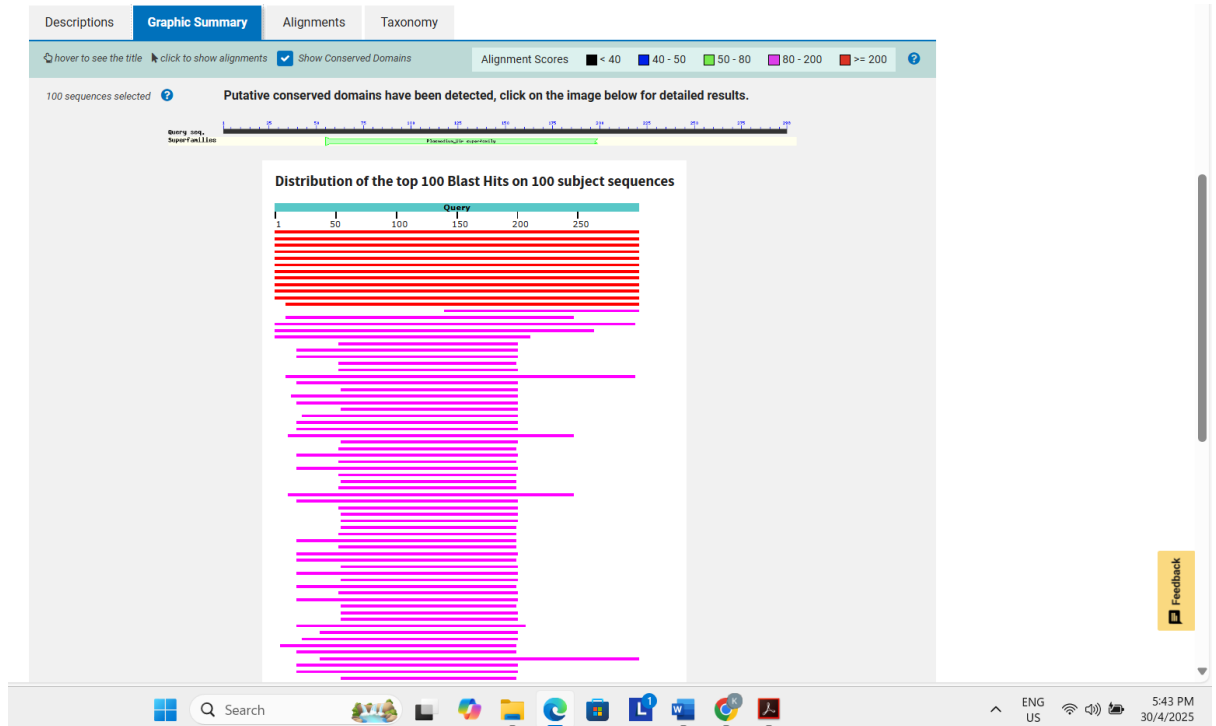
BLAST

Search database nr using DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)
 ☐ Show results in a new window

Algorithm parameters

Here is the DELTA-BLAST result:

6



Compare these results against the new Clustered nr database BLAST

Descriptions **Graphic Summary** Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

100 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Sequences with E-value BETTER than threshold

☒ select all 100 sequences selected

PSI-BLAST iteration 1

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒ vir1 (Plasmodium vivax)	Plasmodium vivax	363	363	100%	2e-126	100.00%	299	CAB86690.1	☒	☒	☒
☒ Plasmodium vivax Vir protein, putative (Plasmodium vivax)	Plasmodium vivax	330	330	100%	2e-113	88.63%	299	SCA82045.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	327	327	100%	4e-112	84.62%	299	CAI7724112.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	314	314	100%	5e-107	85.28%	298	CAI7719847.1	☒	☒	☒
☒ VIR protein (Plasmodium vivax)	Plasmodium vivax	309	309	100%	6e-105	80.27%	299	SCA60288.1	☒	☒	☒
☒ VIR protein (Plasmodium vivax)	Plasmodium vivax	291	291	100%	4e-98	82.27%	299	SCA60058.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	284	284	100%	3e-95	83.61%	299	VUZ93672.1	☒	☒	☒
☒ unnamed protein product (Plasmodium vivax)	Plasmodium vivax	270	270	100%	1e-89	66.22%	299	CAG9473090.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	266	266	100%	3e-88	72.58%	298	VUZ99642.1	☒	☒	☒
☒ unnamed protein product (Plasmodium vivax)	Plasmodium vivax	265	265	100%	6e-88	69.90%	299	CAG9485322.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	247	247	100%	4e-80	58.38%	334	VUZ99734.1	☒	☒	☒
☒ hypothetical protein PVNG_01538 (Plasmodium vivax North Korean)	Plasmodium vivax North Korean	218	218	97%	2e-69	57.84%	307	KNA02384.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	196	196	54%	4e-62	71.79%	195	VUZ99735.1	☒	☒	☒
☒ hypothetical protein PVNG_05816 (Plasmodium vivax Mauritania II)	Plasmodium vivax Mauritania II	194	194	79%	6e-61	83.47%	244	KMZ95672.1	☒	☒	☒
☒ VIR protein (Plasmodium vivax)	Plasmodium vivax	150	150	99%	4e-43	40.74%	301	SCA81956.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	138	138	88%	7e-39	40.30%	264	VUZ99426.1	☒	☒	☒
☒ hypothetical protein PVNG_06431, partial (Plasmodium vivax India VII)	Plasmodium vivax India VII	132	132	70%	3e-37	65.24%	210	KMZ77281.1	☒	☒	☒
☒ hypothetical protein PVNG_05217 (Plasmodium vivax North Korean)	Plasmodium vivax North Korean	125	125	49%	3e-32	23.12%	516	KNA01516.1	☒	☒	☒
☒ PIR protein (Plasmodium vivax)	Plasmodium vivax	124	124	61%	4e-32	14.29%	391	VUZ97285.1	☒	☒	☒
☒ unnamed protein product (Plasmodium vivax)	Plasmodium vivax	124	124	61%	4e-32	16.84%	405	CAG9473105.1	☒	☒	☒
☒ VIR protein, partial (Plasmodium vivax)	Plasmodium vivax	124	124	49%	7e-32	22.64%	455	SCA60244.1	☒	☒	☒

Feedback

5:43 PM 30/4/2025

In a way the results of this particular search are in some sense surprising. BLASTP successfully found a series of matches with very low E values. Whereas DELTA-BLAST successfully found many more matches (using its PSSM-based approach). However its best match (2e-126) does not have as low an Expect value as produced by

BLASTP. In many other DELTA-BLAST searches the E values are better (lower) than those found by BLASTP. The result shown here follows from the PSSM approach used by DELTA-BLAST in which a large set of vir1-related proteins were aligned and scored to define the general properties of this family.

3) Are there globins in fungi?

- i) Perform a PSI-BLAST search using human beta globin (NP_000509) as a query, restricting the output to sequences from fungi (taxid:4751) in the nr database.
- ii) What is the approximate range of lengths of fungal proteins having globin domains? What non-globin domains are often present in fungal globins? Does the presence of these unrelated domains lead to corruption? Why or why not?
- iii) In the first iteration there are several hits (with the E values below the 0.005 threshold). After several more iterations there are many dozens of hits including flavohemoproteins that include a globin domain. These fungal proteins have globin domains that are more related to bacterial than vertebrate orthologs. Most of the fungal flavohemoproteins are quite long (over 400 amino acids and sometimes about 1000 amino acids long), having multiple domains. However, only the globin domain is used for the continued PSI-BLAST iterations.

Lets perform a PSI-BLAST search using human beta globin (NP_000509) as a query, restricting the output to sequences from fungi (taxid:4751) in the nr database.

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blastn **blastp** blastx tblastn tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query, more...

[Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), g(i)s, or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

NP_000509 From To

Or, upload file Choose File No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Standard databases (nr etc.): ☐ Experimental databases

☒ Non-redundant protein sequences (nr) [?](#)

Organism ☐ exclude [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm ☐ Quick BLASTP (Accelerated protein-protein BLAST) ☐ blastp (protein-protein BLAST) ☒ PSI-BLAST (Position-Specific Iterated BLAST) ☐ PHI-BLAST (Pattern Hit Initiated BLAST) ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST Search database nr using PSI-BLAST (Position-Specific Iterated BLAST)

☐ Show results in a new window

[+ Algorithm parameters](#)

[Feedback](#)

The result that there are indeed many globins in fungi. Here show the PSI-BLAST results:

Descriptions **Graphic Summary** Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 500 [?](#)

78 sequences selected [GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

Sequences with E-value BETTER than threshold

☒ select all 24 sequences selected PSI-BLAST iteration 1

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒ globin-like protein [Lipomyces starkeyi]	Lipomyce...	52.8	52.8	73%	3e-06	28.70%	159	KAK9482522.1	☒		
☒ globin-like protein, partial [Lipomyces oligosphaera]	Lipomyce...	50.8	50.8	62%	1e-05	29.79%	150	XP_066777890.1	☒		
☒ globin-like protein [Lipomyces mesembrius]	Lipomyce...	50.1	50.1	94%	3e-05	26.03%	159	KAK9386670.1	☒		
☒ globin-like protein [Lipomyces starkeyi]	Lipomyce...	50.1	50.1	65%	3e-05	29.17%	159	KAK9317215.1	☒		
☒ globin-like protein [Lipomyces kononenkoae]	Lipomyce...	50.4	50.4	90%	5e-05	25.17%	251	KAK9388883.1	☒		
☒ globin-like protein [Lipomyces doorenjongii]	Lipomyce...	49.3	49.3	94%	6e-05	27.40%	159	XP_066798308.1	☒		
☒ globin-like protein [Lipomyces tetrasporus]	Lipomyce...	50.1	50.1	96%	6e-05	23.53%	242	KAK9253566.1	☒		
☒ globin-like protein [Lipomyces starkeyi]	Lipomyce...	50.1	50.1	96%	8e-05	23.45%	310	KAK9484317.1	☒		
☒ globin-like protein [Lipomyces starkeyi]	Lipomyce...	49.7	49.7	96%	1e-04	23.45%	310	KAK9318327.1	☒		
☒ uncharacterized protein BZA70DRAFT_281565 [Myxozyma...]	Myxozym...	49.3	49.3	62%	2e-04	29.79%	375	XP_064767168.1	☒		
☒ globin-like protein [Lipomyces chichuensis]	Lipomyce...	49.3	49.3	96%	2e-04	22.07%	304	XP_066750989.1	☒		
☒ globin-like protein [Lipomyces orientalis]	Lipomyce...	47.8	47.8	94%	2e-04	26.71%	159	KAK9319351.1	☒		
☒ globin-like protein [Lipomyces starkeyi]	Lipomyce...	48.9	48.9	96%	3e-04	23.45%	310	KAK9328966.1	☒		
☒ globin-like protein [Lipomyces doorenjongii]	Lipomyce...	48.5	48.5	96%	3e-04	23.45%	313	XP_066794559.1	☒		
☒ globin-like protein [Lipomyces doorenjongii]	Lipomyce...	48.5	48.5	96%	3e-04	23.45%	315	KAK9428377.1	☒		
☒ globin-like protein [Lipomyces doorenjongii]	Lipomyce...	48.5	48.5	96%	3e-04	23.45%	315	KAK9354001.1	☒		
☒ putative globin-like protein [Yarrowia sp. E02]	Yarrowia...	47.8	47.8	97%	5e-04	27.03%	388	KAG5358934.1	☒		
☒ globin-like protein [Lipomyces oligosphaera]	Lipomyce...	47.4	47.4	62%	8e-04	30.85%	391	XP_066775857.1	☒		
☒ hypothetical protein WICPJ_009049 [Wickerhamomyces p...]	Wickerha...	47.0	47.0	95%	0.001	23.97%	357	KAH3676518.1	☒		
☒ putative globin-like protein [Yarrowia sp. B02]	Yarrowia...	47.0	47.0	97%	0.001	27.03%	385	KAG5358645.1	☒		
☒ globin-like protein [Lipomyces starkeyi]	Lipomyce...	45.1	45.1	65%	0.002	28.12%	159	KAK9327387.1	☒		

[Feedback](#)

https://blast.ncbi.nlm.nih.gov/Blast.cgi?CMD=Ge...

ENG US 5:53 PM 30/4/2025

To see the lengths of these matches inspect the pairwise outputs. You can also reformat to view (or export) the results as a table. Most fungal proteins have lengths of 250 to 450 amino acids. Some are larger, e.g.:

The screenshot displays a BLAST search interface with the 'Alignments' tab selected. It shows two search results for 'globin-like protein' from *Lipomyces* species. Each result includes a download button, GenPept and Graphics links, and navigation controls. The first result, KAK9482522.1, has a length of 159 and one match. The second result, XP_066777890.1, has a length of 150 and one match. Both results show sequence alignments with scores, expect values, and identities. The first alignment has a score of 52.8 bits (125) and an expect value of 3e-06. The second alignment has a score of 50.8 bits (120) and an expect value of 1e-05. Both alignments show high identity percentages (29% and 30% respectively) and positive scores (55/108 and 45/94 respectively). The first alignment also shows gaps (12/108 or 11%). The second alignment shows gaps (3/94 or 3%).

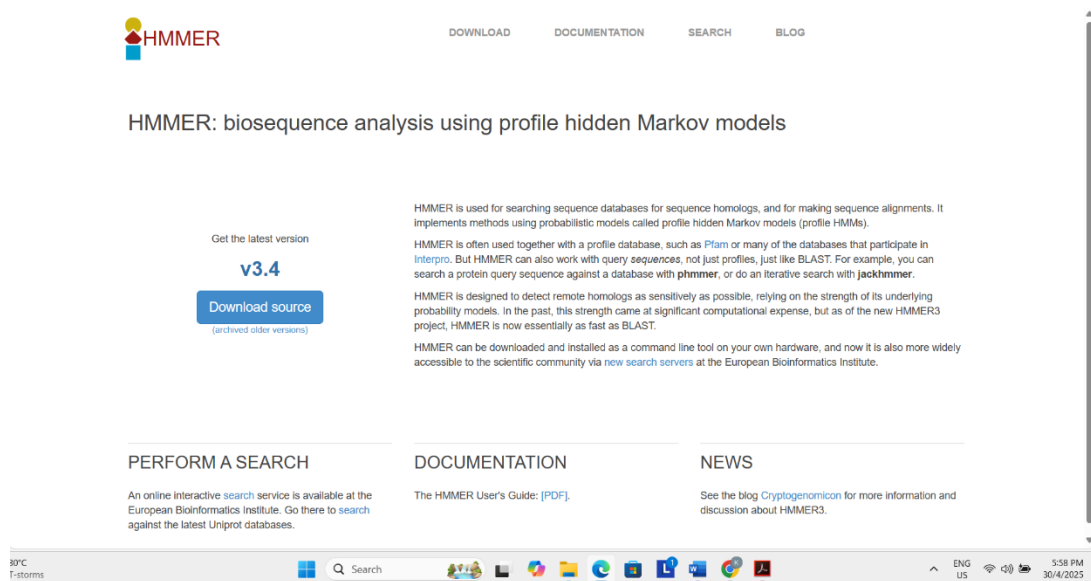
In many cases these larger fungal globins contain repeating globin units. Most fungal proteins have lengths of 250 to 450 amino acids. S globin are often present in fungal globins. Presence of unrelated domains can lead to corruption which result in the high scores for homologous to the unrelated domains. This corruption can then lead to high scores for the assignment and statistical significance to biological-incorrect relationship.

4) Perform HMMER searches

- i) First make two different HMMs. You can obtain sets of vertebrate globin and bacterial/fungal/vertebrate globin sequences as web documents 5.6 and 5.7. The multiple sequence alignments that we use as input to HMMER are in these documents.
- ii) When the profile HMM was built from a multiple sequence alignment of vertebrate alpha and beta globins and used to search the human RefSeq database, there were many database matches, including myoglobin that we could not detect with BLASTP. In contrast, when an alignment of bacterial and fungal globins was used to generate a

profile HMM, the output consisted of one result with a non-significant expect value. Combining several human globins with the bacterial and fungal globins in a multiple sequence alignment resulted in the creation of an HMM that readily detected human globins. Thus, the profile HMM is a model that is sensitive to the choice of sequences that are used as input for the multiple sequence alignment.

- iii) The full results of the HMMER searches for (1) vertebrate, (2) bacterial plus fungal, and (3) bacterial plus fungal plus vertebrate globins are shown in web documents 5.8, 5.9, and 5.10. The HMM match to human myoglobin had a higher score and lower E value in search (3) than in (1).
- iv) HMMer searches are run locally. This search was run against all human RefSeq proteins. You can download NCBI databases such as RefSeq by visiting the file transfer protocol (FTP) site from the home page of NCBI or going directly to ► <http://www.ncbi.nlm.nih.gov/Ftp/> (WebLink 5.1). Place the downloaded database into the same directory as your input sequences for HMMER.



Download hmmer software

Create a new directory (mkdir hmmer)

Change the directory to hmmer: cd hmmer

Move the hmmer package in the download directory to this new created directory :

```
mv ~/Downloads/hmm* hmmer
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\user> mkdir hmmer
mkdir : An item with the specified name C:\Users\user\hmmer already exists.
At line:1 char:1
+ mkdir hmmer
+ ~~~~~
+ CategoryInfo          : ResourceExists: (C:\Users\user\hmmer:String) [New-Item], IOException
+ FullyQualifiedErrorId : DirectoryExist,Microsoft.PowerShell.Commands.NewItemCommand

PS C:\Users\user> cd hmmer
PS C:\Users\user\hmmer> mv ~/Downloads/hmm* hmmer
PS C:\Users\user\hmmer> |
```

- 5) We previously performed a series of BLAST searches using HIV-1 pol as a query (NP_057849).
- i) Perform a BLASTP search using this query. Look at the taxonomy report to see which viruses match this query.
 - ii) Next, repeat the search using several iterations of PSI-BLAST. Compare this taxonomy report to that of the BLASTP search.
 - iii) What do you observe? Are there any nonviral sequences detected in the PSI-BLAST search?
 - iv) Did you expect to find any?

BLASTP searches using HIV-1 pol as a query (NP_057849).

BLAST® » blastp suite Home Recent Results Saved Strategies Help

Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query, more...

[Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

NP_057849 From To

Or, upload file No file chosen [?](#)

Job Title [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Standard databases (nr etc.) ☐ Experimental databases

RefSeq Select proteins (refseq_select) [?](#)

Organism ☐ exclude [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm ☒ blastp (protein-protein BLAST) ☐ PSI-BLAST (Position-Specific Iterated BLAST) ☐ PHI-BLAST (Pattern Hit Initiated BLAST) ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST Search database RefSeq Select proteins (refseq_select) using Blastp (protein-protein BLAST) ☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with +

[sign](#) [+ Algorithm parameters](#) [Feedback](#)

This shows the taxonomy report for BLASTP for NP_057849.

Descriptions **Graphic Summary** Alignments **Taxonomy**

Reports **Lineage** Organism Taxonomy

77 sequences selected [?](#)

Organism	Blast Name	Score	Number of Hits	Description
cellular organisms			82	
• Bacteria	bacteria		33	
• • Actinomycetes	high G+C Gram-positive bacteria		10	
• • • Pseudonocardiaaceae	high G+C Gram-positive bacteria		2	
• • • • Saccharothrix mutabilis	high G+C Gram-positive bacteria	371	1	Saccharothrix mutabilis hits
• • • • Pseudonocardia pini	high G+C Gram-positive bacteria	47.4	1	Pseudonocardia pini hits
• • • • Kribbella italica	high G+C Gram-positive bacteria	50.1	1	Kribbella italica hits
• • • • Corynebacterium senegalense	high G+C Gram-positive bacteria	48.5	2	Corynebacterium senegalense hits
• • • • Corynebacterium marquesiae	high G+C Gram-positive bacteria	48.9	1	Corynebacterium marquesiae hits
• • • • Marmoricola endophyticus	high G+C Gram-positive bacteria	46.6	1	Marmoricola endophyticus hits
• • • • Kribbella koreensis	high G+C Gram-positive bacteria	46.2	1	Kribbella koreensis hits
• • • • Kineosporia babensis	high G+C Gram-positive bacteria	47.0	1	Kineosporia babensis hits
• • • • Mycobacterium innocens	high G+C Gram-positive bacteria	48.1	1	Mycobacterium innocens hits
• • • • Geopseudomonas aromaticivorans	g-proteobacteria	244	3	Geopseudomonas aromaticivorans hits
• • • • Polyangium jinanense	bacteria	55.8	1	Polyangium jinanense hits
• • • • Devosia insulae	a-proteobacteria	52.8	1	Devosia insulae hits
• • • • Deinococcus peraridilitoris	bacteria	51.6	1	Deinococcus peraridilitoris hits
• • • • Eikenella corrodens	b-proteobacteria	52.8	1	Eikenella corrodens hits
• • • • Saccharospirillum mangrovi	g-proteobacteria	50.1	1	Saccharospirillum mangrovi hits
• • • • Deinococcus apachensis	bacteria	49.7	1	Deinococcus apachensis hits
• • • • Nitrococcus mobilis	g-proteobacteria	48.1	1	Nitrococcus mobilis hits
• • • • Hydrogenophilus thiooxidans	proteobacteria	48.1	1	Hydrogenophilus thiooxidans hits
• • • • Deinococcus aerius	bacteria	48.5	1	Deinococcus aerius hits
• • • • Gammaproteobacteria	g-proteobacteria	47.8	1	Gammaproteobacteria hits

[Feedback](#)

PSI-BLAST searches using HIV-1 pol as a query (NP_057849).

BLAST® » blastp suite

Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. more...

Reset page Bookmark

Enter Query Sequence

Enter accession number(s), gI(s), or FASTA sequence(s) [Clear](#)

NP_057849

Query subrange

From To

Or, upload file [Choose File](#) | No file chosen

Job Title

NP_057849 Gag-Pol [Human immunodeficiency

Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database

☒ Standard databases (nr etc.) ☐ Experimental databases

[RefSeq Select proteins \(refseq_select\)](#)

Organism

Optional

Enter organism name or id-completions will be suggested

Exclude

Optional

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ blastp (protein-protein BLAST)

☒ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

BLAST

Search database RefSeq Select proteins (refseq_select) using PSI-BLAST (Position-Specific Iterated BLAST)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign

+ Algorithm parameters

FOLLOW NCBI

30°C
f-storms

Search

ENG
US

6:22 PM
30/4/2025

This shows the taxonomy report for PSI-BLAST for NP_057849.

Number of sequences 100 Run

Descriptions Graphic Summary Alignments **Taxonomy**

Reports Lineage Organism Taxonomy

100 sequences selected

Organism	Blast Name	Score	Number of Hits	Description
cellular organisms			105	
• Bacteria	bacteria		58	
• • Bacillati	bacteria		3	
• • • Actinomycetes	high G+C Gram-positive bacteria		2	
• • • • Saccharothrix mutabilis	high G+C Gram-positive bacteria	371	1	Saccharothrix mutabilis hits
• • • • Kribbella italica	high G+C Gram-positive bacteria	50.1	1	Kribbella italica hits
• • • Paenibacillus	firmicutes	60.5	1	Paenibacillus hits
• Geopseudomonas aromaticivorans	g-proteobacteria	244	3	Geopseudomonas aromaticivora hits
• Wolbachia endosymbiont of Psyllodes chrysocephala	a-proteobacteria	75.1	2	Wolbachia endosymbiont of Psyllodes chrysocephala hits
• • Polyangium spumosum	bacteria	60.5	1	Polyangium spumosum hits
• • Polyangium mundeleinium	bacteria	60.1	1	Polyangium mundeleinium hits
• • Polyangium fumosum	bacteria	60.1	1	Polyangium fumosum hits
• • Polyangium solediatum	bacteria	60.1	1	Polyangium solediatum hits
• • Acetobacter fabarum	a-proteobacteria	60.8	5	Acetobacter fabarum hits
• • Chondromyces crocatus	bacteria	58.9	1	Chondromyces crocatus hits
• • Acetobacter tropicalis	a-proteobacteria	60.1	2	Acetobacter tropicalis hits
• • Acetobacter oeni	a-proteobacteria	60.1	1	Acetobacter oeni hits
• • Sandaracinus amylolyticus	bacteria	58.9	1	Sandaracinus amylolyticus hits
• • Polyangium aurulentum	bacteria	58.5	1	Polyangium aurulentum hits
• • Acetobacter musti	a-proteobacteria	58.5	1	Acetobacter musti hits
• • Chondromyces apiculatus	bacteria	56.6	1	Chondromyces apiculatus hits
• • Cohaesibacter celericrescens	a-proteobacteria	53.9	1	Cohaesibacter celericrescens hits
• • Sorangium atrum	bacteria	55.8	1	Sorangium atrum hits

ENG
US

6:33 PM
30/4/2025

DELTA-BLAST searches using HIV-1 pol as a query (NP_057849).

BLAST [®] » blastp suite

Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. more...

Reset page Bookmark

Enter Query Sequence

Enter accession number(s), g(i)s, or FASTA sequence(s) [?] Clear

NP_057849

Query subrange [?]

From To

Or, upload file

Choose File No file chosen [?]

Job Title

NP_057849 Gag-Pol [Human Immunodeficiency]

Enter a descriptive title for your BLAST search [?]

☐ Align two or more sequences [?]

Choose Search Set

Database

☒ Standard databases (nr etc.) ☐ Experimental databases

RefSeq Select proteins (refseq select) [?]

Organism

Optional

Enter organism name or id—completions will be suggested

Exclude

Optional

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☒ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?]

BLAST

Search database RefSeq Select proteins (refseq_select) using DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with [?] sign

+ Algorithm parameters

FOLLOW NCBI

18°C

Storms tonight

Search

ENG US

6:33 PM

30/4/2025

This shows the taxonomy report for DELTA-BLAST for NP_057849.

Descriptions Graphic Summary Alignments **Taxonomy**

Number of sequences 500 Run

Reports Lineage Organism Taxonomy

500 sequences selected [?]

Organism	Blast Name	Score	Number of Hits	Description
cellular organisms			503	
• Bacteria	bacteria		476	
• • Pseudomonadati	bacteria		465	
• • • Pseudomonadota	proteobacteria		464	
• • • • Gammaproteobacteria	g-proteobacteria		1	
• • • • • Pseudomonadales	g-proteobacteria		23	
• • • • • • Geopseudomonas aromaticivorans	g-proteobacteria	474	1	Geopseudomonas aromaticivorans
• • • • • • • Tamilnaduibacter salinus	g-proteobacteria	172	1	Tamilnaduibacter salinus hits
• • • • • • • Marinobacter zhanjiangensis	g-proteobacteria	171	1	Marinobacter zhanjiangensis hits
• • • • • • • Marinobacter laciisalsi	g-proteobacteria	170	1	Marinobacter laciisalsi hits
• • • • • • • Pseudohongiella nitratreducens	g-proteobacteria	169	1	Pseudohongiella nitratreducens
• • • • • • • Marinobacter	g-proteobacteria	168	6	Marinobacter hits
• • • • • • • Marinobacter vulgaris	g-proteobacteria	168	1	Marinobacter vulgaris hits
• • • • • • • Marinobacter orientalis	g-proteobacteria	167	1	Marinobacter orientalis hits
• • • • • • • Marinobacter bryozoorum	g-proteobacteria	167	1	Marinobacter bryozoorum hits
• • • • • • • Marinobacter subterrani	g-proteobacteria	167	1	Marinobacter subterrani hits
• • • • • • • Marinobacter halodurans	g-proteobacteria	166	1	Marinobacter halodurans hits
• • • • • • • Marinobacter guineae	g-proteobacteria	166	1	Marinobacter guineae hits
• • • • • • • Marinobacter azerbaijanicus	g-proteobacteria	165	1	Marinobacter azerbaijanicus hits
• • • • • • • Marinobacter litoralis	g-proteobacteria	165	1	Marinobacter litoralis hits
• • • • • • • Marinobacter nanhaiticus	g-proteobacteria	165	1	Marinobacter nanhaiticus hits
• • • • • • • Marinobacter confluentis	g-proteobacteria	164	1	Marinobacter confluentis hits
• • • • • • • Marinobacter maroccanus	g-proteobacteria	164	1	Marinobacter maroccanus hits
• • • • • • • Pseudohongiella spirulinae	g-proteobacteria	164	1	Pseudohongiella spirulinae hits

ENG US

6:45 PM

30/4/2025

From the results show above, there are more hits with DELTA-BLAST than BLASTP and PSI-BLAST. There are both viral and non-viral detected in the PSI search. This is expected because the PSSM-based approach is far more sensitive than that of standard BLAST.

- 6) Explore PHI-BLAST using human RBP4 (NP_006735) as a query, restricting the output to bacteria and the RefSeq database.
 - i) Use the PHI pattern `GXW[YF]X[VILMAFY]A[RKH]`. Perform this search, and save the results.
 - ii) Then repeat the search using the PHI pattern `GXW[YF][EA][IVLM]`.
 - iii) How do the results differ?
 - iv) Select one protein that appears as a bacterial protein in a pairwise alignment with the human RBP4 query; what are the E values, and why do they differ?

Here is the set-up of the PHI-BLAST using the PHI pattern `GXW[YF]X[VILMAFY]A[RKH]` search:

The screenshot shows the NIH BLAST suite interface. The top navigation bar includes the NIH logo and links for Home, Recent Results, Saved Strategies, and Help. The main section is titled "Standard Protein BLAST". Under "Enter Query Sequence", the accession number "NP_006735" is entered. The "Job Title" is "NP_006735 retinol-binding protein 4 isoform". In the "Choose Search Set" section, "Standard databases (nr etc.)" is selected, and "Reference proteins (refseq, protein)" is chosen. The "Organism" is set to "Bacteria (taxid 2)". In the "Program Selection" section, "PHI-BLAST (Pattern Hit Initiated BLAST)" is selected, and the PHI pattern "GXW[YF]X[VILMAFY]A[RKH]" is entered. The "BLAST" button is visible at the bottom left of the form.

For every database match, the results include asterisks indicating where this pattern matches. And the top two patterns result for PHI-BLAST using the PHI pattern `GXW[YF]X[VILMAFY]A[RKH]` are shown here:

Descriptions Graphic Summary **Alignments** Taxonomy Number of sequences: 500

Alignment view: Pairwise [Restore defaults](#) Download

500 sequences selected

[Download](#) [GenPept](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

MULTISPECIES: lipocalin family protein [unclassified Tamiana]
Sequence ID: [WP_303422675.1](#) Length: 179 Number of Matches: 1

Range 1: 6 to 111 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Identities	Positives	Gaps
31.4 bits(90)	2e-05	32/108(30%)	55/108(50%)	2/108(1%)

Pattern: *****
Query 3: WNWALLLAALGSGRAERDCRVSSFRVKNFDFKARFSGTWYMAKKDPEGLFQQNIVAE 62
Sbjct 6: WWSLLTSLGLLSFLSRTIPEKAVAVK-SFDKAKYLKGYEIALDFKYKDLNNTTAE 64

Query 63: FSVDETGHSAATAGRVRLNNMDVCAHMGFTDTEDPAKFKMKYNG 110
Sbjct 65: YSLNNGTIKVNQGFDTISKQNEQSIGK-AKFVDEDDVAKLVKVSFFG 111

[Download](#) [GenPept](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

lipocalin family protein [Sphingomonas sp.]
Sequence ID: [WP_331027152.1](#) Length: 186 Number of Matches: 1

Range 1: 45 to 173 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Identities	Positives	Gaps
31.0 bits(89)	2e-05	41/146(28%)	70/146(47%)	19/146(13%)

Pattern: *****
Query 34: DKARFSGTWYMAKKDPEGLFQQNIVAEFSVDETGHSAATAGRVRLNNMDVCAHMG 93
Sbjct 45: DLARYAGKWFELARHENRFQGLDAVTAEYTLKDGKRLNLNSG-IRAGEKERDVAEGHA 103

Query 94: TFDTEPAKFKMKYNGVASFLQKGNDDHIVD--TDYTYAVQVSCRLNLNDGTCADSY 151
Sbjct 104: IVADEETNAKLLKLSFGP---LYTG--DYNVLDRGDDYMSIV-----GEPSGRY 148

Query 152: SFVSRDPDPLPPEAKIVRQRQEL 177
Sbjct 149: LNWLSRDPHP-APETLQHLRERVAL 173

[Download](#) [GenPept](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

lipocalin family protein [Sphingomonas tagetis]

Then repeat the PHI-BLAST using the PHI pattern GXW[YF][EA][IVLM] search:

NIH National Library of Medicine National Center for Biotechnology Information Log in

BLAST® » blastp suite Home Recent Results Saved Strategies Help

blastn **blastp** blastx tblastn tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more...

Enter Query Sequence

Enter accession number(s), g(i)s, or FASTA sequence(s) [Clear](#)

NP_006735

Query subrange: From To

Or, upload file: [Choose File](#) No file chosen

Job Title: NP_006735: retinol-binding protein 4 isoform...
Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database: ☒ Standard databases (nr etc.) ☐ Experimental databases

RefSeq Select proteins (refseq_select)

Organism: [Add organism](#)

Exclude: ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm: ☐ blastp (protein-protein BLAST) ☐ PSI-BLAST (Position-Specific Iterated BLAST) ☒ PHI-BLAST (Pattern Hit Initiated BLAST)

GxW[YF][EA][IVLM]

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

BLAST Search database RefSeq Select proteins (refseq_select) using PHI-BLAST (Pattern Hit Initiated BLAST)

☐ Show results in a new window

The top two results are identical, except that there are 6 asterisks now instead of 8 asterisks previously. Here the result for PHI-BLAST using the PHI pattern GXW[YF][EA][IVLM]:

Descriptions Graphic Summary **Alignments** Taxonomy

Number of sequences 500 **Run**

Alignment view Pairwise [Restore defaults](#) Download

420 sequences selected

Download [GenPept](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

retinol-binding protein 4 isoform a precursor [Homo sapiens]
Sequence ID: [NP_006735.2](#) Length: 201 Number of Matches: 1

Range 1: 1 to 201 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

	Score	Expect	Identities	Positives	Gaps
	399 bits(1043)	6e-116	201/201(100%)	201/201(100%)	0/201(0%)

Pattern Query 1 MKWMALLLLAALGSGRAEDRCVSSFRVKNFDFKARFSGTNYAMAKDPEGLFLQDNIV 60
Sbjct 1 MKWMALLLLAALGSGRAEDRCVSSFRVKNFDFKARFSGTNYAMAKDPEGLFLQDNIV 60
Pattern

Query 61 AEFSDVETGQMSATAGRVRLLNNNDVCDMVGTFDTEDPAKFKMKYGVASF LQKGN 120
Sbjct 61 AEFSDVETGQMSATAGRVRLLNNNDVCDMVGTFDTEDPAKFKMKYGVASF LQKGN 120

Query 121 DHWIVDTYDVTYAVQYSCRLNLNLDGTCADSYFVFSRDPNGLPPEAQKIVRQREELCLA 180
Sbjct 121 DHWIVDTYDVTYAVQYSCRLNLNLDGTCADSYFVFSRDPNGLPPEAQKIVRQREELCLA 180

Query 181 RQYRLIVHNGYCDGRSERNLL 201
Sbjct 181 RQYRLIVHNGYCDGRSERNLL 201

Related Information
[Gene](#) - associated gene details
[AlphaFold Structure](#) - 3D structure displays
[Genome Data Viewer](#) - aligned genomic context

Download [GenPept](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

retinol-binding protein 4 isoform 2 precursor [Mus musculus]
Sequence ID: [NP_035385.1](#) Length: 201 Number of Matches: 1

Range 1: 1 to 201 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

	Score	Expect	Identities	Positives	Gaps
	352 bits(922)	8e-102	172/201(86%)	185/201(92%)	0/201(0%)

Pattern Query 1 MKWMALLLLAALGSGRAEDRCVSSFRVKNFDFKARFSGTNYAMAKDPEGLFLQDNIV 60
Sbjct 1 MHWIAL+LLAALG AERDCRVSSFRVKNFDFKARFSG NYA+AKKDPEGLFLQDNIV 60
Pattern

Query 61 AEFSDVETGQMSATAGRVRLLNNNDVCDMVGTFDTEDPAKFKMKYGVASF LQKGN 120
Sbjct 61 AEFSDVETGQMSATAGRVRLLNNNDVCDMVGTFDTEDPAKFKMKYGVASF LQKGN 120

Related Information
[Gene](#) - associated gene details
[AlphaFold Structure](#) - 3D structure displays
[Genome Data Viewer](#) - aligned genomic context

Repeat the PHI-BLAST without the PHI pattern search:

BLAST® » blastp suite Home Recent Results Saved Strategies Help

blastn **blastp** blastx tblastn tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more...

Enter Query Sequence [Reset page](#) [Bookmark](#)

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) Query subrange [From](#) [To](#)

NP_006735

Or, upload file [Choose File](#) No file chosen

Job Title [NP_006735 retinol-binding protein 4 isoform...](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Standard databases (nr etc.) ☐ Experimental databases

☒ RefSeq Select proteins (refseq_select) [?](#)

Organism [Enter organism name or id - completions will be suggested](#) ☐ exclude [Add organism](#)

Optional [Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown](#) [?](#)

Exclude ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WPI) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm ☐ blastp (protein-protein BLAST) ☐ PSI-BLAST (Position-Specific Iterated BLAST) ☒ PHI-BLAST (Pattern Hit Initiated BLAST)

[Enter a PHI pattern](#) [?](#)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST) [Choose a BLAST algorithm](#) [?](#)

BLAST Search database RefSeq Select proteins (refseq_select) using PHI-BLAST (Pattern Hit Initiated BLAST) ☐ Show results in a new window

Here the result for PHI-BLAST without using the PHI pattern:

Alignment view Pairwise Restore defaults Download

27 sequences selected

[Download](#) [GenPept](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

retinol-binding protein 4 isoform a precursor [Homo sapiens]
Sequence ID: [NP_006735.2](#) Length: 201 Number of Matches: 1

Range: 1: 1 to 201 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
420 bits(1080)	3e-149	Compositional matrix adjust.	201/201(100%)	201/201(100%)	0/201(0%)
Query 1	MKWMAALLLLAALGSGRAERDCRVSSFRVKNFDKARFSGTNYAMAKKDPEGLFLQDNIV				60
Sbjct 1	MKWMAALLLLAALGSGRAERDCRVSSFRVKNFDKARFSGTNYAMAKKDPEGLFLQDNIV				60
Query 61	AEFSVDETEQMSATAKGRVRLINNNDVCADPWGTFDTEDPAKFKMYMGVASFQKGN				120
Sbjct 61	AEFSVDETEQMSATAKGRVRLINNNDVCADPWGTFDTEDPAKFKMYMGVASFQKGN				120
Query 121	DHWIVDTDYDTYAVQYSCRLNLDGTCADSYSFVSRDPNGLPPEAQIVRQREELCLA				180
Sbjct 121	DHWIVDTDYDTYAVQYSCRLNLDGTCADSYSFVSRDPNGLPPEAQIVRQREELCLA				180
Query 181	RQYRLIVHNGYCDGRSERNLL 201				
Sbjct 181	RQYRLIVHNGYCDGRSERNLL 201				

[Related Information](#)
[Gene](#) - associated gene details
[AlphaFold Structure](#) - 3D structure displays
[Genome Data Viewer](#) - aligned genomic context

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retinol-binding protein 4 isoform 2 precursor [Mus musculus]
Sequence ID: [NP_035385.1](#) Length: 201 Number of Matches: 1

Range: 1: 1 to 201 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
369 bits(947)	5e-129	Compositional matrix adjust.	172/201(86%)	185/201(92%)	0/201(0%)
Query 1	MKWMAALLLLAALGSGRAERDCRVSSFRVKNFDKARFSGTNYAMAKKDPEGLFLQDNIV				60
Sbjct 1	MHWMAALLLLAALGSGRAERDCRVSSFRVKNFDKARFSGTNYAMAKKDPEGLFLQDNII				60
Query 61	AEFSVDETEQMSATAKGRVRLINNNDVCADPWGTFDTEDPAKFKMYMGVASFQKGN				120
Sbjct 61	AEFSVDETEQMSATAKGRVRLINNNDVCADPWGTFDTEDPAKFKMYMGVASFQKGN				120
Query 121	DHWIVDTDYDTYAVQYSCRLNLDGTCADSYSFVSRDPNGLPPEAQIVRQREELCLA				180
Sbjct 121	DHWIVDTDYDTYAVQYSCRLNLDGTCADSYSFVSRDPNGLPPEAQIVRQREELCLA				180
Query 181	RQYRLIVHNGYCDGRSERNLL 201				
Sbjct 181	RQYRLIVHNGYCDGRSERNLL 201				

[Related Information](#)
[Gene](#) - associated gene details
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[Genome Data Viewer](#) - aligned genomic context

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Inspecting the results from both searches, the first one (with the longer PHI pattern) had more database matches that were significant (above threshold). If we instead do a PHI-BLAST search without using a PHI pattern, only 27 match is significant. Therefore a PHI pattern can dramatically improve the sensitivity of a BLAST protein search.

Protein that appears as a bacterial protein in a pairwise alignment with the human RBP4 query: Select seq ref[[NP_006735.2](#)] retinol-binding protein 4 isoform a precursor [Homo sapiens]

E values:

PHI- BLAST pattern GXW[YF]X[VILMAFY]A[RKH] – 9e-116

PHI- BLAST pattern GXW[YF][EA][IVLM] – 6e-116

PHI- BLAST without pattern – 3e-149

Inspecting the results from three searches, the first one (with the longer PHI pattern) had more database matches that were significant (above threshold). If we instead do a PHI-BLAST search without using a PHI pattern, only 27 match is significant. Therefore a PHI pattern can dramatically improve the sensitivity of a BLAST protein search